Tizen, etc). An HTML5 mobile app is a Web application developed with that version of the Web content standard and designed for smartphones, tablets and other handheld devices. The earlier versions couldn't support the complex functionalities required for developing mobile applications. HTML5 makes it effortless to create a fully featured Web application that can be updated remotely with new functionality. As these applications are run from the Web rather than stored locally, the user is not required to download updates to view or use the app. With the rise in mobile device usage and mobile technologies, it has become a herculean task for developers to focus on one particular platform such as Android, iPhone or Windows. HTML5 apps eradicate this problem as they are examples of cross-platform development. HTML5 can create apps that are compatible not only with mobile devices but also desktop and notebook browsers, for an unvarying experience across all user devices.

Key features of mobile devices

Offline support: Offline support includes application cache, Web storage and indexed database APIs that store HTML, JavaScript, CSS and media resources, locally. Cache is used to create Web based applications that work even if the user is not connected to the Internet — for example, JavaScript calculator, Canvas based games, etc. To provide this support, a manifest file should be created which specifies the application's resources.

```
CACHE MANIFEST
# Version 0.1
offline.html
/iui/iui.js
/iui/iui.css
/iui/loading.gif
/iui/toolbar.png
/iui/Button1.png
/images/books.jpg
/images/author.png
/images/bookcover.jpg
```

The 'manifest' attribute should specify the URL of the manifest file, as follows:

```
<html manifest="path/filename">
```

Multimedia: HTML5 apps have advanced capabilities for streaming video and audio data, handling graphics and animation. They also add semantic elements, form controls and multimedia components. To support multimedia, HTML5 has two tags, audio and video. A simple video tag is shown below:

```
<video src="myvideo.mp4" controls />
```

And to support different browsers:

```
<video poster="myvideo.jpg" controls>
    <source src="myvideo.m4v" type="video/mp4" />
    <source src="myvideo.ogg" type="video/ogg" />
    <embed src="/to/my/video/player"></embed>
</video>
```

Geolocation API: The geolocation API helps users to share their location with the websites. This is not actually a part of HTML5, but is implemented in JavaScript. It makes use of the available information of the longitude and latitude on the JavaScript page and sends it to the remote server.

A simple example is as follows:

```
<script>
var z = document.getElementById("example");
function getLocation() {
     if (navigator.geolocation) {

navigator.geolocation.getCurrentPosition(showPosition);

} else {
       z.innerHTML = "Geolocation is not supported by this
browser.";
     }
}

function showPosition(position) {
z.innerHTML = "Latitude:  " + position.coords.latitude +
"<br>Longitude: "+ position.coords.longitude;
}
</script>
```

Canvas: Canvas is particularly interesting since it facilitates the use of graphics without the need for any plugins or other technologies, other than JavaScript and CSS. It can be used to make interesting charts, graphs, images, arcs, etc. Transformations can also be performed on it.

Example: You can set up a canvas measuring the same size as the screen, as follows:

```
     canvas = document.getElementById('mycanvas');

     // Set canvas dimensions
     canvas.width = window.innerWidth;
     canvas.height = window.innerHeight;
```

Frameworks for application development

There are umpteen frameworks for HTML5 application development. Here are a few of them.

JQuery Mobile: This is a unified user interface system across all popular mobile device platforms and is built on jQuery and jQuery UI.